

RepoMind AI - Autonomous Software Engineering Intelligence Platform

Vision

Build an autonomous AI platform that understands GitHub repositories without requiring users to clone them locally. The system investigates repositories like a senior engineer by planning, gathering evidence, and answering complex engineering questions.

Problem Statement

Developers spend significant time understanding unfamiliar repositories, reading documentation, tracing code, reviewing pull requests, and learning architecture. Existing tools mostly provide repository chat. RepoMind focuses on investigation, planning, reasoning, and actionable engineering outputs.

Target Users

Software Engineers, Open Source Contributors, Students, Engineering Managers, Tech Leads, AI Engineers.

Tech Stack

Frontend: Next.js, React, Tailwind CSS, Shadcn UI, React Flow.

Backend: FastAPI.

AI: LangGraph + OpenAI/Anthropic/OpenRouter.

Protocol: MCP.

Storage: PostgreSQL (Neo4j optional).

Observability: LangSmith.

Deployment: Docker.

Core Workflow

User provides a GitHub URL. Planner Agent understands the goal, Repository Explorer gathers repository metadata and files through GitHub APIs/MCP, specialized agents investigate architecture, pull requests, issues, dependencies and history, Reflection Agent decides whether more evidence is needed, Synthesizer generates the final answer.

Agents

Planner Agent, Repository Explorer, File Reader, Pull Request Agent, Issue Agent, Git History Agent, Architecture Agent, Dependency Agent, Security Agent, Performance Agent, Refactoring Agent, Documentation Agent, Reflection Agent, Synthesizer Agent.

Major Features

1. Repository Overview (frameworks, languages, architecture).
2. Natural language repository Q&A.;
3. Pull Request analysis and summaries.
4. Issue investigation.
5. Architecture and dependency visualization.
6. Feature implementation planning.
7. Impact analysis for code changes.
8. Repository health report.
9. Developer onboarding guide generation.

10. Release note generation from PRs.

Agentic Capabilities

- Dynamic planning for every query.
- Intelligent tool selection using MCP.
- Reflection and self-evaluation.
- Confidence scoring.
- Memory of previously explored files.
- Parallel specialist agents.
- Investigation until sufficient evidence is collected.

Example Queries

- Explain authentication flow.
- Summarize PR #28.
- Why was Redis introduced?
- What breaks if JWT middleware is removed?
- Which files should I edit to add Google OAuth?
- Generate onboarding documentation.
- Compare two repositories.

Advanced Roadmap

Phase 1: GitHub integration and repository explorer.

Phase 2: Planner and repository chat.

Phase 3: PR/Issue analysis.

Phase 4: Architecture graphs.

Phase 5: Impact analysis and implementation planner.

Phase 6: Multi-agent collaboration and reflection.

Phase 7: Code generation, PR creation, and autonomous engineering assistance.

Future Extensions

Support GitLab, Bitbucket and Azure DevOps, integrate terminal execution, CI/CD awareness, deployment analysis, automated code review, security audits, and autonomous pull request creation.

Why It Stands Out

This project demonstrates planning, multi-agent orchestration, MCP integration, GitHub automation, software architecture analysis, reasoning over engineering artifacts, and production-style AI workflows rather than a simple chatbot.